

Analyzing Bipartite Plant-Animal Networks

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Introduction

Like most networks, food webs contain various patterns which can provide clues about how and why nodes interact with each other on both a global and local scale. Although extensive research has been poured into analyzing various sets of completely observed food webs, these generally focus on the global, trophic patterns of predator-prey interactions. In specifically examining the bipartite interactions of both plant-pollinator and plant-frugivore networks it is believed a greater understanding of the coevolutionary process between these species can be developed. Initial research into the topic seemed to suggest these mutualistic networks had a small percentage of species acting as “super-generalists” or nodes with high edge wealth, while the majority of nodes had much lower edge wealth. This seemed to closely fit a power-law or truncated power-law distribution, where a change in one quantity will create a proportional change in another quantity (“Power Law and Power Law Distribution” 1999). From this analysis, conclusions were drawn that these networks must have a great robustness also and, hence, a high degree of resilience to the removal of less connected nodes (Jordano *et al.* 2003). However, more recent research in the fields of ecology harshly contradicts this analysis, arguing for a phenomenon called trophic cascade. The idea of trophic cascades claims food webs, despite appearing robust, actually have high levels of sensitivity to changes in the network, and the absence of one seemingly lightly connected species can cause extreme fluctuations in the populations of nodes, including non-neighbor nodes. For example, the removal of the Grey Wolf from Yellowstone’s food web resulted in a reduction in the aspen tree population, despite the Grey Wolf having a relatively light edge weight compared to the super-generalist species of the network (Beschta *et al.* 2023). Again however, most of this research seems to focus on the larger predator-prey relationships, but it still remains important to analyze the specifics of plant-animal networks and coevolutionary processes they describe. This paper aims to reexamine initial research done in the paper *Invariant properties in coevolutionary networks of plant-animal interactions* (2003) by Pedro Jordano, Jordi Bascompte, and Jens Olesen on mutualistic plant-animal networks using more modern tools for analyzing power-law like distributions. Reasking “(1) if there are shared patterns in the topology of plant-mutualistic networks, irrespective of whether they are pollination or seed dispersal networks? (2) What are the implications and biological causes for these patterns?” (Jordano *et al.* 2003). Setting forth these questions with a different hypothesis, which predicts power-law and truncated power-law will not actually produce the best fit for these mutualistic networks.

Methods

To begin the process, data was collected from existing and available sets of plant-animal networks. To recreate the original paper and test its conclusions, 2 plant-pollinator and 2 plant-frugivore networks listed in the appendix as fitting power-law distributions were selected to be tested. It should be noted some networks listed in the original paper were specified as not fitting the power-law, so I have been sure to only pull from networks they specified did in fact seem to follow a power-law or truncated power-law distribution. Given much of this data was collected over 30 years ago, it is important to note that it may not necessarily provide an accurate representation of the current circumstances of the environments and networks from which these samples were collected. In an attempt to remedy this 2 additional, more modern networks, one plant-pollinator and one plant-frugivore, which were not examined in the original study were also included.

In order to best recreate the original paper most mathematical methods were reused, starting with the creation of an adjacency matrix R which describes the interactions between plant i and animal j , where p and a are contained within the set of all plants P and all animals A in the network, respectively. as a

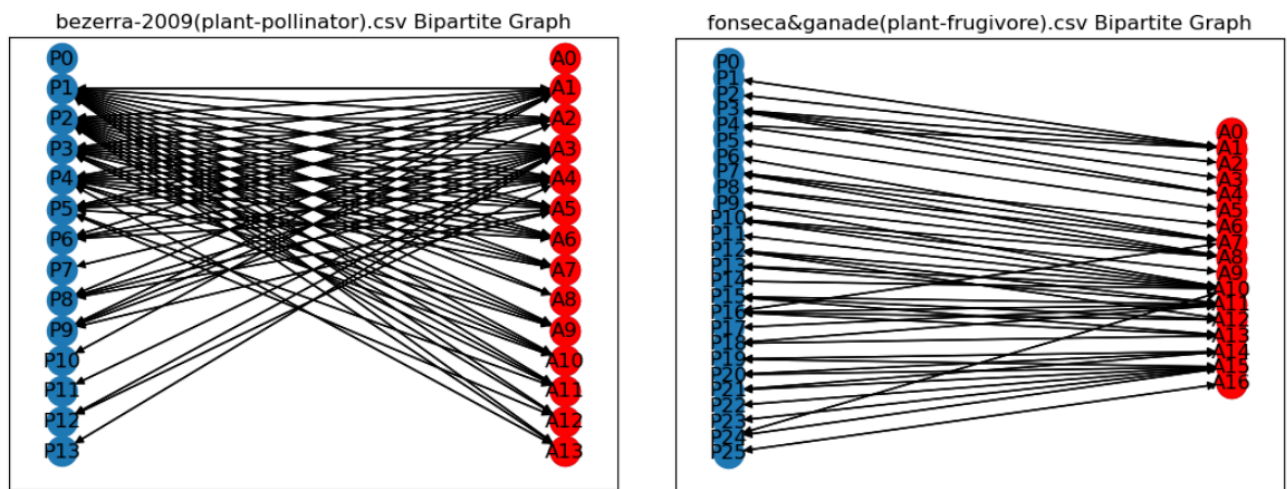
binary relationship of 1 if they have any kind of contact, whether pollination based or trophic, and 0 otherwise. The edge wealth of any node i , represented as k_i , was determined as the number of non-zero elements in row or column representing node i 's interactions.

$$\mathbf{R} = [a_{ij}]_{A \times P}$$

$$\text{where } a_{ij} = \begin{cases} 1, & \text{if } P_i \text{ and } A_j \text{ interact} \\ 0, & \text{otherwise} \end{cases}$$

Figures from *Invariant properties in coevolutionary networks of plant-animal interactions*(2003) p. 70-71

Using networkx package in python(Appendix 1), the adjacency matrices for each network were constructed first into directed bipartite graphs as done in the paper, and then undirected bipartite graphs in attempt to test if this had any effects on the fit of edge wealth distribution to the power-law curve.



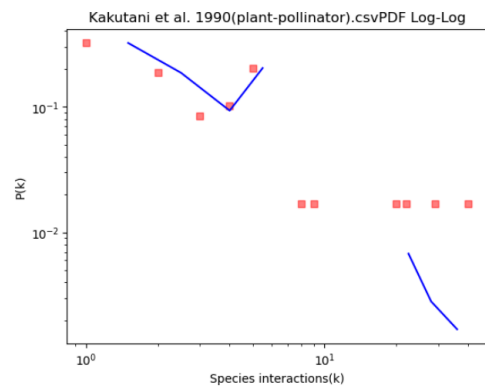
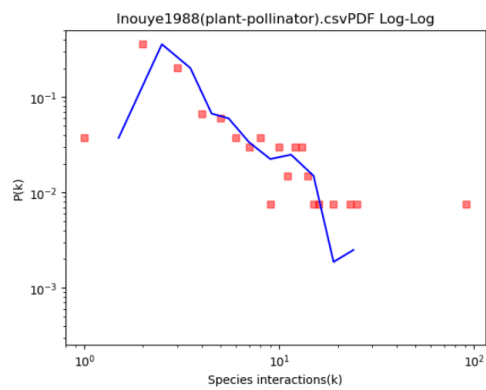
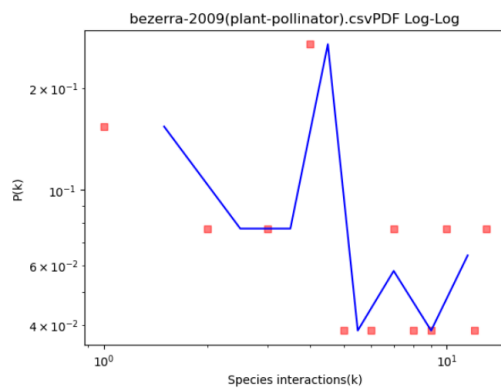
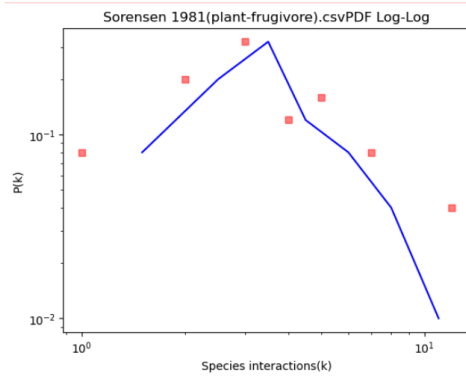
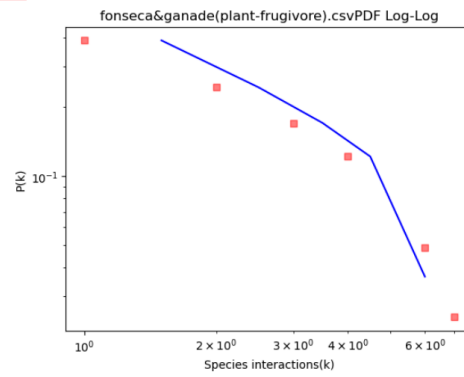
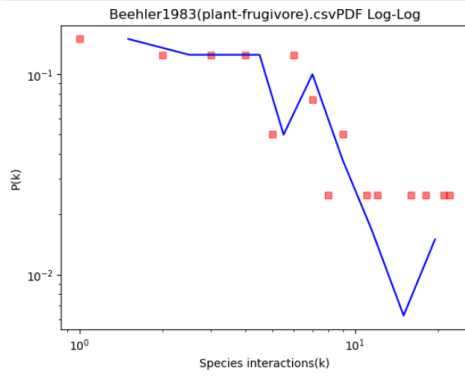
Graphs constructed from data found on Index of Complex Networks; blue represents plants in the network red represents animals

Next, I attempted to fit each network to the power-law curve, where the curve could be calculated as $\Pr(k) = k^{-\gamma}$ for all values of k_i ; where γ is a fitted constant. In the original paper it is unclear exactly how this constant was calculated to fit each network; however, they do determined γ to be with the range of 1.23 $\pm$ 0.04 for plant-pollinator networks and 1.12 $\pm$ 0.13 for plant-frugivore networks(Jordano *et al.* 2003). To determine gamma and how well it fit the given graph I used *power-law* package(Appendix 2), which uses methods developed in *Power-Law Distributions In Empirical Data*(Clauset *et al.* 2009) to calculate a γ for a provided network. Once a γ had been calculated for each network, I plotted the probability density function(PDF) for each network in an attempt to fit the power-law curve to these values. Finally, I again used python to plot the cumulative distribution function(CDF) as $\Pr(K \geq k)$ against the degree, k , for each of the networks. This was to check if the network somewhat followed the rules of a power-law distribution. It was expected that if the CDF curve was above the mean degree slope for almost all nodes this had the potential of being a power law distribution, since most nodes in a power-law distribution should have a significantly high $\Pr(K \geq k)$.

Results

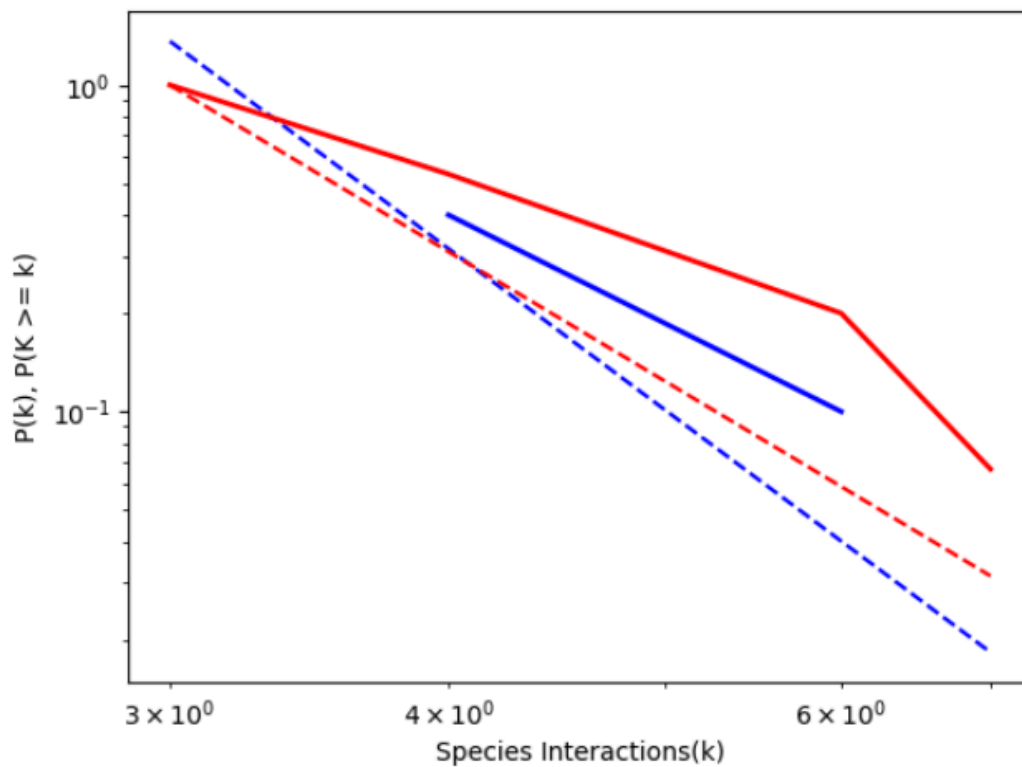
Name	Nodes (n)	Edges (m)	Avg Degree ($\langle k \rangle$)	Fitted Constant (γ)	Power-law fit vs Exp (p-val)	Power-law fit vs Log-normal (p-val)
Beehler 1983	42	238	11.33	3.269	0.39	0.78
Fonseca 2008	43	96	4.47	5.085	0.29	0.03
Soresn 1981	28	94	6.71	4.006	0.21	0.80
Plant-frugivore Avg	37.6	142.7	7.5	4.120		
Bezerra 2009	28	142	10.14	3.108	0.88	0.83
Inouye 1988	134	744	11.10	2.483	0.01	0.26
Kakutani 1990	61	268	8.79	3.037	0.16	0.54
Plant-pollinator Avg	74.3	384.7	10.01	2.876		

Overall, the plant-frugivore networks had an average γ of about 4.120, while the plant-pollinators had a smaller γ of around 2.876. Based on the research done on power-law, if a natural occurring network is does truly fit a power-law distribution well it typically will have $2 < \gamma < 3$ (Clauset *et al.* 2009). Additionally, when compared to other potential distributions, frequently it could not be determined if the power law truly was a better fit with p-values frequently exceeding the acceptable confidence threshold of 0.05. In the two cases where the power-law package comparison tool produced a comparison with p-values within the acceptable limit, in the Fonseca 2008 it was determined that the log-normal distribution was actually a slightly better fit (Only by 1%). Only for one network, Inouye 1988, could confidently be determined as best-fit by the power-law.

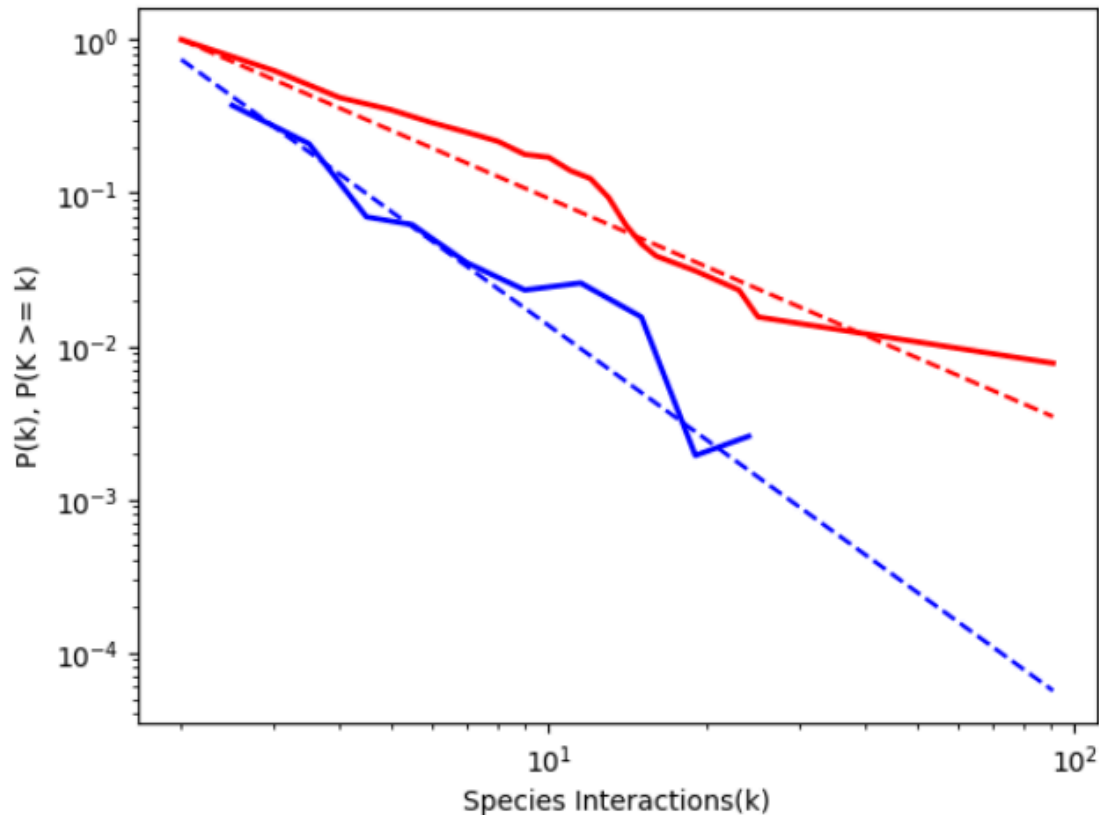


These PDF graphs demonstrate how in most cases the power-law curve did not or could not fit itself to the network's edges. Only in a Fonseca 2008 and Inouye 1998, can good power-law fits be noticeably observed.

Examining the CDF's of the Fonseca 2008 and Inouye 1998, it can be observed the edge distribution could possibly be a power-law distribution as for the most part the CDF curve is above mean edge slope. Again for the poorly fitting networks this cannot be observed in their CDFs.



*Figure above is CDF for Fosenca 2008 network



*Figure above is CDF for Inouye 1998 network

Discussion

Although significantly less data was compiled and tested, based on the findings of this paper it appears the original hypothesis claiming power-law was usually a good fit for most mutualistic networks does not hold true. With only two out of six networks even coming close to fitting the power-law distribution, and only one having the power-law as a best fit it must be reconsidered whether or not the original conclusions about the robustness of these plant-animal networks was accurate. Responding to the focus question of this paper, it is now unclear whether or not shared patterns in the topology of plant-mutualistic networks exist. Of course attempts have not been made to fit all possible distributions to these networks, but it does appear some networks fit the distribution and others do not. This indicates a variation in the topology of these networks, which may result from the phenology of ecosystems in which they exist. The biological causes for absence of these patterns, cannot be answered with this data; however, absence of these patterns implies a possible lack of robustness within these networks. This would support the current ecological findings, which argue many ecosystems are in fact very sensitive to species/node loss. By increasing the pool of networks in the future, we may be able to affirm these claims from a networking perspective. Additionally, it is important to consider the opinions of ecologists on what it means to have a robust ecosystem may differ from the mathematical claims to having a robust network. Network analysts may look at the loss of the Grey Wolf in Yellowstone and determine the network generally survived well enough in its absence, as opposed to the ecologist who believes the reintroduction of the wolves has greatly improved the biological stability of the network (Beschta *et al.* 2023). Overall, the power-distribution can fit itself well to many distributions, but it does not appear to fit this data as

well as initial thought. From this, the paper concludes the hypothesis to accurately reflect the results of these trials.

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Appendix

Entry 1:

```
def bipartite_from_cvs(filename):
    bezerra = np.genfromtxt(filename, delimiter=',')
    rows = bezerra.shape[0]
    cols = bezerra.shape[1]

    G = nx.DiGraph()
    nodes = []
    z = {}

    #Pull out nodes
    for plant in range(rows):
        p_name = "P" + str(plant)
        nodes.append(p_name)
        z[p_name] = 0

    for animal in range(cols):
        a_name = "A" + str(animal)
        nodes.append(a_name)
        z[a_name] = 1

    G.add_nodes_from(nodes)

    #Add edges
    for plant in range(rows):
        p_name = "P" + str(plant)
        for animal in range(cols):
            a_name = "A" + str(animal)

            if bezerra[plant][animal] > 0:
                #print("Plant:", p_name, " Animal:", a_name, " edge:", bezerra[plant][animal])
                G.add_edge(p_name, a_name, weight=1)
                G.add_edge(a_name, p_name, weight=1)

    #Draw the graph the way I want it to look
    pos = {}
    plants = []
    animals = []
    for i, node in enumerate(nodes):
        if z[node] == 0:
            pos[node] = (0, -i)
            plants.append(node)
        else:
            pos[node] = (1, -i + len(nodes) / 2)
            animals.append(node)

    #print(plants)
    #print(animals)

    nx.draw_networkx(G, pos=pos, with_labels=True)
    nx.draw_networkx_nodes(G, pos=pos, nodelist=animals, node_color='red')
    plt.title(filename + " Bipartite Graph")
    plt.show()

    return G, z
```

Entry 2:

```
#Generate data
def power_law(G):
    k = []
    for node in G.nodes:
        k_i = len(list(G.neighbors(node)))
        #print(k_i)
        if k_i > 0:
            k.append(k_i)

    #print("K:", sorted(k))
    data = np.array(k) # data can be list or numpy array
    results = powerlaw.Fit(data)
    gamma = results.power_law.alpha
    k_min = results.power_law.xmin
    print("Gamma:", gamma)
    print("k_min:", k_min)
    R, p = results.distribution_compare('power_law', 'lognormal')
    print("Power-law vs Log-normal:", R, p)
    R, p = results.distribution_compare('power_law', 'exponential')
    print("Power-law vs Exponential:", R, p)

    return results, k

def plot_power_law(k, fit, name):
    # input : a list of degrees, e.g., from a networkx graph G
    # output: a plot of the PDF of the degree distribution Pr(k) on Log-Log axes for k>=1

    kmax = max(k)

    # histogram and plot
    counts, bins = np.histogram(k, bins=[i for i in range(kmax+2)], density=True)
    #plt.semilogx(bins[1:-1], counts[1:], 'rs-', alpha=0.5)
    #plt.show()
    plt.loglog(bins[1:-1], counts[1:], 'rs', alpha=0.5)
    plt.xlabel('Degree, k')
    plt.ylabel('Pr(k)')
    plt.title(name + "PDF Log-Log")

    # Plot the data
    figPDF = powerlaw.plot_pdf(k, color='b')
    figPDF.set_ylabel("P(k)")
    figPDF.set_xlabel(r"Species Interactions(k)")
    figname = 'FigPDF'
    plt.show()

    return

def plot_pdf_v_cdf(fit):
    fig = fit.plot_pdf(color='b', linewidth=2)
    fit.power_law.plot_pdf(color='b', linestyle='--', ax=fig)
    fit.plot_ccdf(color='r', linewidth=2, ax=fig)
    fit.power_law.plot_ccdf(color='r', linestyle='--', ax=fig)
    fig.set_ylabel("P(k), P(K >= k)")
    fig.set_xlabel(r"Species Interactions(k)")
    plt.show()

    return
```